



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (INFORMATION TECHNOLOGY) EXAMINATIONS - JAN/FEB
2023

SUBJECT : ICT 2154 - DIGITAL SYSTEMS
(MAKEUP)

DIGITAL SYSTEMS [ICT 2154]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Design a 4 - bit Carry Look Ahead binary adder using logic gates. (5)
- 1B) Design an asynchronous UP counter to count from 2 to 6 using positive edge triggered D - flip flops and minimum external gates. (3)
- 1C) Design T - flip flop using NOR latch and external gates. (2)
- 2A) Design a sequence detector using Moore model with one input Y and one output Z. The output Z is HIGH whenever the sequence "101" is detected, otherwise the output is LOW. Overlapping of the sequence is allowed. Implement using JK- flip flops and minimum number of external gates. (5)
- 2B) Realize the following Boolean functions using suitable ROM
 $F_1(A,B,C) = \overline{A}C + \overline{B}C + AB$ and $F_2(A,B,C) = ABC + \overline{B}C + \overline{A}$ (3)
- 2C) Design a combinational circuit that compares two, 2 - bit numbers to check if $A \geq B$ using logic gates. The circuit output is equal to 1, if the two numbers satisfy the condition $A \geq B$. (2)
- 3A) Design a code converter to convert a decimal digit represented in excess 3 to 8 4 -2 -1 code using 3:8 decoders and minimum external gates. (5)
- 3B) Design full subtractor using 4:1 MUXs ONLY. (3)
- 3C) Using synchronous hexadecimal counter ICs and external gates, design a 2 - digit hexadecimal down counter which counts from 94H to 35H and repeats. (2)
- 4A) Minimize the following using Tabulation Method and realize using basic gates.
 $Y(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$ (5)
- 4B) Design a sequence generator to generate the sequence 1101011 by ring counter using JK Flip flop and external gates. (3)
- 4C) Design and implement a full adder circuit using a 3:8 decoder. (2)
- 5A) Design synchronous divide by 10 counter using D-flip flops. (5)

- 5B) Design a 3X2 bit combinational multiplier using basic gates. (3)
- 5C) Solve the SoP function minimization of 5 variable K-Map using the following expression: (2)
- $$F(A, B, C, D, E) = \sum(0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31).$$

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